Gayatri Vidya Parishad College of Engineering Autonomous

Computer Science and Engineering | Course Code: CS201

Data Structures and Algorithms - Advanced Data Structures & Algorithmic Ana

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Overview: Complete syllabus guide: Red-Black trees, Disjoint-set union, Dijkstra and Bellman-Fo

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Algorithmic Complexity & Linear Structures
Big-O, Omega, Theta notations, Master Theorem, Stacks, Queues, Doubly linked lists, Circular buffers

Unit 2: Balanced Search Trees & Heaps
Binary Search Trees, AVL rotations, Red-Black tree properties, Binary heaps, HeapSort, Priority queu

Unit 3: Graph Traversal & Spanning Trees
Breadth-First Search, Depth-First Search, Topological sort, Prim and Kruskal Minimum Spanning Tree a

Unit 4: Shortest Paths & Dynamic Programming
Dijkstra single-source shortest path, Bellman-Ford, Floyd-Warshall, 0/1 Knapsack, Longest Common Sub

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Solve the recurrence relation Tn = 3Tn/2 + On log n using Master Theorem and verify with recursion tre
2. Insert the keys [15, 20, 24, 10, 13, 7, 30, 36] into an empty AVL tree and show intermediate rotation
3. Formulate the dynamic programming recurrence and write the bottom-up table construction for the 0/1 Kn

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest - Introduction to Algorithms MIT Press 3r
* Mark Allen Weiss - Data Structures and Algorithm Analysis in C++ Pearson 4th Ed

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